SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Security Incident Mapping (SIM)

Course: SEN202

Project Title: Security Incident Mapping (SIM)

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1. Introduction

1.1 Purpose

The Security Incident Mapping (SIM) system is a web-based application developed to improve security within the campus community. It enables students, staff, and landlords to report security incidents quickly while providing administrators with a centralized dashboard to monitor and manage reported incidents. The aim is to encourage prompt reporting, improve emergency response, and support informed decision-making for campus security management.

1.2 Scope

The system allows members of the campus community to report security incidents through a simple web interface. It stores incident reports, enables administrators to monitor and manage reports, and provides basic statistics on reported incidents. The application is designed as a prototype for academic purposes and demonstrates the fundamental features of a digital crime reporting platform.

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1.3 Objectives

The objectives of the Security Incident Mapping system are:

- To provide an easy and fast method for reporting security incidents.
- To reduce delays in reporting crimes within the campus environment.
- To assist administrators in monitoring reported incidents.
- To improve campus security through digital reporting.
- To replace manual reporting with an efficient electronic system.

2. Overall Description

2.1 Product Perspective

The Security Incident Mapping (SIM) system is a standalone web application developed using HTML, CSS, and JavaScript. The application stores information using the browser's Local Storage and does not require a server or online database. It serves as a prototype that demonstrates the basic operations of a security incident reporting system.

2.2 Product Functions

The system provides the following functions:

- Submit new security incident reports.

- Store reports locally in the browser.
- Display all submitted reports.
- Search reports by category or location.
- Filter reports based on incident category.
- Update the status of reported incidents.
- Delete reports.
- Display dashboard statistics.

2.3 User Classes

Student

Students can:

- Report security incidents.
- Upload supporting evidence.
- View safety information. • Track submitted reports.

Staff

Staff members can:

- Report security incidents.
- Upload supporting evidence.

- Track submitted reports.

Landlord

Landlords can:

- Report incidents occurring around student residences.
- Upload supporting evidence.
- Track submitted reports.

Administrator

The administrator is responsible for:

- Logging into the system.
- Viewing all incident reports.
- Searching reports.
- Filtering reports.
- Updating report status.
- Deleting reports.
- Viewing dashboard statistics.

3. Functional Requirements

FR1 – Home Page

The system shall display a homepage containing:

- Navigation menu
- About section
- Features section
- Statistics section
- Report Incident button

FR2 – Incident Reporting

The system shall allow users to submit incident reports containing:

- Full Name
- Anonymous reporting option
- Email Address
- Phone Number
- Incident Category
- Date
- Time
- Location

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- Severity Level
- Description of the incident

FR3 – Data Storage

The system shall store submitted reports using the browser's Local Storage.

FR4 – Dashboard

The administrator dashboard shall display:

- Total reports
- Pending reports
- Resolved reports
- Theft cases
- Recent incident records

FR5 – Search Function

The administrator shall be able to search reports using:

- Incident location
- Incident category

FR6 – Filter Function

The administrator shall be able to filter reports according to the selected incident category.

FR7 – Report Status

The administrator shall be able to update the status of reports from Pending to Resolved.

FR8 – Delete Report

The administrator shall be able to permanently delete incident reports.

4. Non-Functional Requirements

Performance

- The system should load within three seconds under normal conditions.
- Dashboard updates should occur immediately after a report is submitted.

Reliability

- Submitted reports should remain available until deleted by the administrator.
- Required fields must be validated before submission.

Security

- User input should be validated.

- The administrator dashboard should only be accessible after authentication in future versions.

Usability

The system should:

- Be simple to learn.
- Provide a user-friendly interface.
- Support desktop and mobile devices.

Maintainability

The source code should be modular, readable, and easy to modify.

Portability

The application should operate correctly on modern web browsers including:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Opera

5. System Design

Input

The system accepts the following information:

- Full Name
- Email Address
- Phone Number
- Incident Category
- Date
- Time
- Location
- Severity Level
- Description of the incident

Processing

The system performs the following operations:

1. The user fills in the incident report form.
2. JavaScript validates the input data.
3. The report is saved to Local Storage.
4. The dashboard retrieves all stored reports.

5. Statistics are automatically generated.
6. The administrator manages the submitted reports.

Output

The system produces:

- Confirmation of successful report submission.
- Dashboard containing all incident reports.
- Statistics of reported incidents.
- Search and filter results.
- Updated report status.
- Deleted report records.

6. Object Listing

Object	Description
User	A student, staff member, or landlord who submits an incident report.
Administrator	The person responsible for managing reports.
	A record containing details of a reported

Incident Report	incident.
Dashboard	Displays reports and system statistics.
Report Form	Collects incident information from users.
Local Storage	Stores reports in the browser.
Search Module	Searches reports based on keywords.
Object	Description
Filter Module	Filters reports according to category.
Statistics Module	Calculates and displays report summaries.

7. Future Improvements

Future versions of the system may include:

- GPS-based incident location tracking.
- Google Maps integration.
- SMS notifications.
- Email notifications.

- AI-powered crime prediction.
- Secure user authentication.
- Cloud database integration.
- Mobile application support.
- Image recognition for uploaded evidence.
- Integration with police and emergency response agencies.

8. Conclusion

The Security Incident Mapping (SIM) system provides an efficient and user-friendly approach to reporting and managing security incidents within a campus community. By allowing students, staff, and landlords to submit reports electronically and enabling administrators to monitor these reports through a centralized dashboard, the system contributes to faster reporting, improved communication, and better security management. Although this prototype uses Local Storage for demonstration purposes, it establishes a solid foundation for future enhancements such as cloud storage, real-time location mapping, user authentication, and integration with emergency response services.

Functional Requirements Summary

Requirement ID	Description
FR1	Display the Home Page
FR2	Submit Incident Reports
FR3	Store Reports in Local Storage
FR4	Display the Dashboard
FR5	Search Reports
Requirement ID	Description
FR6	Filter Reports
FR7	Update Report Status
FR8	Delete Reports

Non-Functional Requirements Summary

Requirement ID	Description
NFR1	Fast system response time
NFR2	Reliable report storage
NFR3	Input validation and basic security

NFR4	User-friendly interface
NFR5	Responsive design
NFR6	Easy maintenance
NFR7	Cross-browser compatibility